

Problem 5.5

Suppose that \$2,500 is deposited on March 8 and withdrawn on October 3 of the same year, and that the interest rate is 5%. Find the amount of interest earned if it is computed using:

- (a) exact simple interest (assume a non-leap year),
- (b) ordinary simple interest,
- (c) Banker's Rule.

We use the simple interest formula:

$$I = Prt,$$

where $P = 2500$, $r = 0.05$, and t is the time in years.

1. Number of Days

From March 8 to October 3, the exact number of days is:

$$23 + 30 + 31 + 30 + 31 + 31 + 30 + 3 = 209 \text{ days.}$$

For ordinary simple interest, 30-day months are used. From March to October there are 7 months, so:

$$7 \cdot 30 - 5 = 205 \text{ days.}$$

(a) Exact Simple Interest

Using the exact number of days and a 365-day year:

$$I = 2500(0.05) \left(\frac{209}{365} \right).$$

Therefore,

$$I \approx 71.58.$$

(b) Ordinary Simple Interest

Using 205 days and a 360-day year:

$$I = 2500(0.05) \left(\frac{205}{360} \right).$$

Thus,

$$I \approx 71.18.$$

(c) Banker's Rule

Using the exact 209 days and a 360-day year:

$$I = 2500(0.05) \left(\frac{209}{360} \right).$$

So,

$$I \approx 72.57.$$